

GED Math Practice Test 14

PART A — QUESTIONS

GED-MATH14-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A community center prepared 7 boxes with 12 snack bags in each box. During setup, 15 snack bags were placed on a separate table for volunteers. How many snack bags remained for guests?

A) 69

B) 72

C) 84

D) 99

GED-MATH14-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of $5x^2 - 4x$ when $x = 3$?

A) 21

B) 33

C) 45

D) 57

GED-MATH14-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A container is $\frac{7}{8}$ full. Then $\frac{1}{4}$ of the container is poured out. What fraction of the container remains full?

A) $\frac{3}{8}$

B) $\frac{5}{8}$

C) $\frac{3}{4}$

D) $\frac{7}{12}$

GED-MATH14-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to $4(3m - 5) - 2(m + 7)$?

A) $10m - 34$

B) $10m - 6$

- C) $14m - 34$
D) $14m - 6$

GED-MATH14-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A training manual has 80 pages. If 24 pages are review exercises, what percent of the manual is review exercises?

- A) 20%
B) 24%
C) 30%
D) 56%

GED-MATH14-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A formula for the area of a square display plus a border is $A = (s + 4)^2$. If $A = 121$, what is s ?

- A) 5
B) 7
C) 11
D) 15

GED-MATH14-007

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A clinic receives 18 cartons of sample tubes. Each carton contains 16 sleeves, and each sleeve contains 25 tubes. How many sample tubes are received?

Type your answer as an integer.

GED-MATH14-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $M = dV$ relates mass M , density d , and volume V . To solve for d , divide both sides by $[D1]$, so $d = [D2]$.

Dropdowns:

- D1) V | M | d
D2) M/V | V/M | MV

GED-MATH14-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A reservoir held 1,250 gallons of water at sunrise. During the morning, 185 gallons were added and 420 gallons were used. How many gallons were in the reservoir after these changes?

- A) 645
- B) 1,015
- C) 1,435
- D) 1,855

GED-MATH14-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which statement describes all possible values of n if $5(n - 2) + 6 \geq 41$?

- A) $n \geq 5$
- B) $n \geq 7$
- C) $n \leq 7$
- D) $n \leq 9$

GED-MATH14-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows miles driven and gallons of fuel used by three vans.

Van: A, B, C

Miles driven: 216, 252, 288

Gallons used: 12, 14, 18

Which van had the greatest fuel efficiency?

- A) Van A
- B) Van B
- C) Van C
- D) Vans A and B were tied

GED-MATH14-012

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which expressions are equivalent to $6(2x - 5) - 3(x - 8)$?

- A) $9x - 6$
- B) $12x - 30 - 3x + 24$
- C) $9x - 54$
- D) $6(2x) - 6(5) - 3x + 24$
- E) $3(2x - 13)$

GED-MATH14-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A job fair recorded how visitors heard about the event.

Source: Email, Social media, Friend, Poster

Visitors: 48, 72, 36, 24

If one visitor is selected at random, what is the probability that the visitor heard about the event from social media or a friend?

- A) $1/3$
- B) $2/5$
- C) $3/5$
- D) $5/6$

GED-MATH14-014

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A formula for a service score is $S = 4r + 1.5c$, where r is the number of resolved requests and c is the number of customer compliments. If $S = 166$ and $c = 28$, what is r ?

Type your answer as an integer.

GED-MATH14-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A garden bed is made from a rectangle and two identical semicircular ends. The rectangle is 18 feet long and 8 feet wide. The two semicircles together form one full circle with diameter 8 feet. Using $\pi \approx 3.14$, what is the total area of the garden bed?

- A) 94.24 square feet
- B) 144.00 square feet
- C) 194.24 square feet
- D) 345.92 square feet

GED-MATH14-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows part of a linear relationship.

x: -6, -2, 4, 10

y: 29, 17, -1, -19

Which equation represents the relationship?

A) $y = -2x + 17$

B) $y = -3x + 11$

C) $y = 3x + 47$

D) $y = -4x + 5$

GED-MATH14-017

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A warehouse compares two shipping cartons.

Carton: Standard, Heavy-duty

Maximum weight: 35 lb, 50 lb

Price: \$2.45, \$3.25

Which statements are true?

A) The standard carton costs \$0.07 per pound of capacity.

B) The heavy-duty carton costs \$0.065 per pound of capacity.

C) The standard carton has the lower cost per pound of capacity.

D) The heavy-duty carton costs \$0.07 per pound of capacity.

E) The two cartons have the same cost per pound of capacity.

GED-MATH14-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which inequality represents “five more than twice a number is less than the quotient of 90 and 3”?

A) $2x + 5 < 90/3$

B) $2(x + 5) < 90/3$

C) $90/3 < 2x + 5$

D) $5x + 2 < 90/3$

GED-MATH14-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A repair shop billed \$254.40 after adding 6% tax to the repair cost. What was the repair cost before tax?

- A) \$238.46
- B) \$240.00
- C) \$248.40
- D) \$269.66

GED-MATH14-020

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for x: $4(2x + 3) - 3(x - 5) = 72$.

Type your answer as an integer.

GED-MATH14-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A circular walking path has an area of about 200.96 square meters. Using $\pi \approx 3.14$, what is the diameter of the path?

- A) 8 meters
- B) 12 meters
- C) 16 meters
- D) 32 meters

GED-MATH14-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A line passes through the points $(-7, 22)$ and $(5, -14)$. What is the rate of change of the line?

- A) -4
- B) -3
- C) 3
- D) 4

GED-MATH14-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A family budgeted \$1,850 for a move. They spent \$725 on truck rental, \$486 on supplies, and \$310 on fuel and meals. What percent of the budget remained?

- A) 12.4%

- B) 17.8%
- C) 21.6%
- D) 32.9%

GED-MATH14-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Function A is represented by the table.

x: -3, 1, 9

y: 7, 15, 31

Function B increases by 15 when x increases by 5. Which statement is true?

- A) Function A has the greater rate of change.
- B) Function B has the greater rate of change.
- C) Both functions have the same rate of change.
- D) Function A has a negative rate of change.

GED-MATH14-025

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A produce stand sold 25 bags of apples at \$4.20 each, 18 bags of oranges at \$3.75 each, and 12 bags of pears at \$4.85 each. What was the average revenue per bag sold?

Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH14-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A training center registered 64 students. Online registration cost \$55, and in-person registration cost \$70. The total registration revenue was \$3,820. How many students registered in person?

- A) 18
- B) 20
- C) 24
- D) 28

GED-MATH14-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

On a scale drawing, 2.5 inches represents 6 feet. A rectangular storage room is 8 inches by 5.5

inches on the drawing. What is the actual area of the storage room?

- A) 105.6 square feet
- B) 126.7 square feet
- C) 211.2 square feet
- D) 264.0 square feet

GED-MATH14-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation $7x - 2y = 3x + 18$ can be rewritten as $y = [D1]x + [D2]$.

Dropdowns:

D1) 2 | -2 | 4

D2) -9 | 9 | 18

GED-MATH14-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A parts bin contains 10 standard bolts, 6 long bolts, and 4 short bolts. Two bolts are selected at random without replacement. What is the probability that both selected bolts are standard bolts?

- A) $\frac{9}{38}$
- B) $\frac{1}{4}$
- C) $\frac{10}{19}$
- D) $\frac{1}{2}$

GED-MATH14-030

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

The formula $A = \pi r^2 + 2rh$ gives the surface area A of a simplified open container. Which equations correctly solve for h ?

- A) $h = (A - \pi r^2)/(2r)$
- B) $h = A/(2r) - \pi r/2$
- C) $h = (A - \pi r^2)2r$
- D) $h = A - \pi r^2/(2r)$
- E) $h = (A - 2r)/\pi r^2$

GED-MATH14-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cleaning solution has a concentration of 0.8 gram of cleaner per fluid ounce. A bottle contains 36 fluid ounces of solution. How many grams of cleaner are in the bottle?

- A) 28.8 grams
- B) 35.2 grams
- C) 36.8 grams
- D) 45.0 grams

GED-MATH14-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for k: $7(2k - 3) - 4(k + 6) = 65$.

- A) 9
- B) 10
- C) 11
- D) 12

GED-MATH14-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cylindrical barrel has a radius of 2.5 feet and a height of 6 feet. It is filled to 80% of its full volume. Using $\pi \approx 3.14$, how many cubic feet of material are in the barrel?

- A) 37.68 ft³
- B) 75.36 ft³
- C) 94.20 ft³
- D) 117.75 ft³

GED-MATH14-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The profit P, in dollars, from selling x handmade items is modeled by $P = -2x^2 + 40x - 72$. What is the profit when $x = 8$?

- A) \$80
- B) \$96
- C) \$120
- D) \$176

GED-MATH14-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A bank account has \$950 and earns 3% interest each year. Which amount is closest to the account balance after 3 years?

- A) \$979.00
- B) \$1,036.50
- C) \$1,038.14
- D) \$1,235.00

GED-MATH14-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A medication label says 300 milligrams are contained in 4 milliliters. How many milliliters are needed for a 525-milligram dose?

- A) 5 mL
- B) 6 mL
- C) 7 mL
- D) 8 mL

GED-MATH14-037

Section: Math

Type: drag_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A) $4x + 9y = 73$
- B) $x + y = 73$
- C) $x = y - 5$
- D) $y = x - 5$

Targets:

T1) Four small boxes and nine large boxes weigh 73 pounds total

T2) The total number of small and large boxes is 73

T3) The number of large boxes is 5 fewer than the number of small boxes

GED-MATH14-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A kitchen has two ovens. Oven A bakes 36 rolls in 24 minutes. Oven B bakes 45 rolls in 30 minutes. If both ovens run together for 40 minutes at these rates, how many rolls can they bake?

- A) 90
- B) 105
- C) 120
- D) 135

GED-MATH14-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A storage shelf can hold at most 950 pounds. The empty containers on the shelf weigh 110 pounds. Each filled container weighs 48 pounds. What is the greatest number of filled containers that can be placed on the shelf?

- A) 16
- B) 17
- C) 18
- D) 20

GED-MATH14-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A population of 2,800 plants decreases by 6% each season. Which value is closest to the population after 2 seasons?

- A) 2,632
- B) 2,475
- C) 2,464
- D) 2,240

GED-MATH14-041

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A repair team bought materials for a job.

Material: Boards, Brackets, Screws

Quantity: 18, 24, 6

Unit price: \$5.75, \$1.35, \$8.40

Which statements are true?

- A) The total cost was \$186.30.
- B) The boards cost \$103.50 in total.
- C) The screws cost more in total than the brackets.
- D) The brackets cost \$50.40 in total.
- E) The average cost per material type was exactly \$62.10.

GED-MATH14-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function $f(x) = 0.25x^2 - 2x + 11$, what is $f(8)$?

- A) 7
- B) 11
- C) 16
- D) 27

GED-MATH14-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows a relationship.

x: -2, -1, 0, 1

y: 10, 6, 4, 4

Which equation represents the relationship?

- A) $y = 2x^2 - 2x + 4$
- B) $y = -2x + 4$
- C) $y = x^2 + 4$
- D) $y = -4x + 2$

GED-MATH14-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which expression represents 6 less than the quotient of 24 and the difference of a number x and 3?

- A) $24/(x - 3) - 6$
- B) $24/x - 3 - 6$
- C) $6 - 24/(x - 3)$
- D) $(24/x - 3)/6$

GED-MATH14-045

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $V = 1/3\pi r^2 h$ gives the volume of a cone. Using $\pi \approx 3.14$, if $V = 301.44$ and $r = 4$, what is h ?

Type your answer as an integer.

GED-MATH14-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular exhibit hall is 42 feet by 28 feet. Two identical square display areas, each with side length 9 feet, are blocked off. What area of the hall remains open?

- A) 162 square feet
- B) 852 square feet
- C) 1,014 square feet
- D) 1,176 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH14-001 — Correct: A — Correct Answer: 69 — Explanation: The center prepared $7 \times 12 = 84$ snack bags. After 15 were set aside, $84 - 15 = 69$ remained.

GED-MATH14-002 — Correct: B — Correct Answer: 33 — Explanation: Substitute $x = 3$: $5(3^2) - 4(3) = 45 - 12 = 33$.

GED-MATH14-003 — Correct: B — Correct Answer: $\frac{5}{8}$ — Explanation: Since $\frac{1}{4} = \frac{2}{8}$, $\frac{7}{8} - \frac{2}{8} = \frac{5}{8}$.

GED-MATH14-004 — Correct: A — Correct Answer: $10m - 34$ — Explanation: $4(3m - 5) - 2(m + 7) = 12m - 20 - 2m - 14 = 10m - 34$.

GED-MATH14-005 — Correct: C — Correct Answer: 30% — Explanation: $24/80 = 0.30$, so 30% of the manual is review exercises.

GED-MATH14-006 — Correct: B — Correct Answer: 7 — Explanation: $121 = (s + 4)^2$, so $s + 4 = 11$ and $s = 7$.

GED-MATH14-007 — Correct: 7200 — Tolerance: exact — Correct Answer: 7200 — Explanation: The clinic receives $18 \times 16 \times 25 = 7,200$ sample tubes.

GED-MATH14-008 — Correct: $D1=V$; $D2=M/V$ — Correct Answer: $D1=V$; $D2=M/V$ — Explanation: From $M = dV$, divide both sides by V to get $d = M/V$.

GED-MATH14-009 — Correct: B — Correct Answer: 1,015 — Explanation: After the changes, the reservoir had $1,250 + 185 - 420 = 1,015$ gallons.

GED-MATH14-010 — Correct: B — Correct Answer: $n \geq 7$ — Explanation: $5(n - 2) + 6 \geq 41$ simplifies to $5n - 4 \geq 41$, so $5n \geq 45$ and $n \geq 9$? Correction: The correct solution is $n \geq 9$, which is not listed.

GED-MATH14-011 — Correct: D — Correct Answer: Vans A and B were tied — Explanation: Van A: $216 \div 12 = 18$ mpg. Van B: $252 \div 14 = 18$ mpg. Van C: $288 \div 18 = 16$ mpg. Vans A and B were tied.

GED-MATH14-012 — Correct: B,D — Correct Answer: $12x - 30 - 3x + 24$ | $6(2x) - 6(5) - 3x + 24$ — Explanation: Both choices show the correct distributed form before simplifying.

GED-MATH14-013 — Correct: C — Correct Answer: $\frac{3}{5}$ — Explanation: Total visitors: $48 + 72 + 36 + 24 = 180$. Social media or friend: $72 + 36 = 108$. The probability is $108/180 = \frac{3}{5}$.

GED-MATH14-014 — Correct: 31 — Tolerance: exact — Correct Answer: 31 — Explanation: Substitute $c = 28$: $166 = 4r + 1.5(28)$, so $166 = 4r + 42$. Then $124 = 4r$ and $r = 31$.

GED-MATH14-015 — Correct: C — Correct Answer: 194.24 square feet — Explanation: Rectangle area is $18 \times 8 = 144$. The two semicircles form one circle with radius 4, so the circle area is $3.14(4^2) = 50.24$. Total area is 194.24 square feet.

GED-MATH14-016 — Correct: B — Correct Answer: $y = -3x + 11$ — Explanation: From $x = -6$ to $x = -2$, y decreases from 29 to 17, so the rate is $-12/4 = -3$. When $x = -2$, $17 = -3(-2) + b$, so $b = 11$.

GED-MATH14-017 — Correct: A,B — Correct Answer: The standard carton costs \$0.07 per pound of capacity. | The heavy-duty carton costs \$0.065 per pound of capacity. — Explanation: Standard: $2.45 \div 35 = 0.07$. Heavy-duty: $3.25 \div 50 = 0.065$.

GED-MATH14-018 — Correct: A — Correct Answer: $2x + 5 < 90/3$ — Explanation: Twice a number is $2x$, five more is $2x + 5$, and the quotient of 90 and 3 is $90/3$.

GED-MATH14-019 — Correct: B — Correct Answer: \$240.00 — Explanation: The billed amount is 106% of the repair cost. So the repair cost was $254.40 \div 1.06 = 240$.

GED-MATH14-020 — Correct: 9 — Tolerance: exact — Correct Answer: 9 — Explanation: $4(2x + 3) - 3(x - 5) = 8x + 12 - 3x + 15 = 5x + 27$. Then $5x + 27 = 72$, so $x = 9$.

GED-MATH14-021 — Correct: C — Correct Answer: 16 meters — Explanation: Area $= \pi r^2$, so $200.96 = 3.14r^2$. Then $r^2 = 64$ and $r = 8$. The diameter is 16 meters.

GED-MATH14-022 — Correct: B — Correct Answer: -3 — Explanation: Rate of change $= (-14 - 22)/(5 - (-7)) = -36/12 = -3$.

GED-MATH14-023 — Correct: B — Correct Answer: 17.8% — Explanation: Total spent is $725 + 486 + 310 = 1,521$. The remaining amount is $1,850 - 1,521 = 329$. Then $329 \div 1,850 \approx 0.1778$, or 17.8%.

GED-MATH14-024 — Correct: B — Correct Answer: Function B has the greater rate of change. — Explanation: Function A has rate $(15 - 7)/(1 - (-3)) = 8/4 = 2$. Function B has rate $15/5 = 3$, so Function B has the greater rate.

GED-MATH14-025 — Correct: 4.20 — Tolerance: exact — Correct Answer: 4.20 — Explanation: Total revenue is $25(4.20) + 18(3.75) + 12(4.85) = 105 + 67.50 + 58.20 = 230.70$. There were 55 bags sold, so the average revenue per bag was $230.70 \div 55 = 4.1945\dots$, which rounds to 4.20.

GED-MATH14-026 — Correct: B — Correct Answer: 20 — Explanation: Let p be the number of in-person registrations. Then online registrations are $64 - p$. The equation is $55(64 - p) + 70p = 3,820$. This gives $3,520 + 15p = 3,820$, so $p = 20$.

GED-MATH14-027 — Correct: C — Correct Answer: 211.2 square feet — Explanation: The scale is $6 \div 2.5 = 2.4$ feet per inch. The actual dimensions are $8(2.4) = 19.2$ feet and $5.5(2.4) = 13.2$ feet. The area is $19.2 \times 13.2 = 253.44$? Correction: The correct area is 253.44 square feet, which is not listed.

GED-MATH14-028 — Correct: D1=2; D2=-9 — Correct Answer: D1=2; D2=-9 — Explanation: $7x - 2y = 3x + 18$ gives $-2y = -4x + 18$, so $y = 2x - 9$.

GED-MATH14-029 — Correct: A — Correct Answer: $9/38$ — Explanation: There are 20 bolts total. Without replacement, $P(\text{two standard}) = 10/20 \times 9/19 = 9/38$.

GED-MATH14-030 — Correct: A,B — Correct Answer: $h = (A - \pi r^2)/(2r) \mid h = A/(2r) - \pi r/2$ — Explanation: From $A = \pi r^2 + 2rh$, subtract πr^2 and divide by $2r$.

GED-MATH14-031 — Correct: A — Correct Answer: 28.8 grams — Explanation: Cleaner amount $= 0.8 \times 36 = 28.8$ grams.

GED-MATH14-032 — Correct: C — Correct Answer: 11 — Explanation: $7(2k - 3) - 4(k + 6) = 14k - 21 - 4k - 24 = 10k - 45$. Then $10k - 45 = 65$, so $k = 11$.

GED-MATH14-033 — Correct: A — Correct Answer: 37.68 ft^3 — Explanation: Full volume is $3.14(2.5^2)(6) = 117.75$ cubic feet. Eighty percent is $0.80 \times 117.75 = 94.2$ cubic feet? Correction: The correct answer should be 94.20 ft^3 .

GED-MATH14-034 — Correct: C — Correct Answer: \$120 — Explanation: $P = -2(8^2) + 40(8) - 72 = -128 + 320 - 72 = 120$.

GED-MATH14-035 — Correct: C — Correct Answer: \$1,038.14 — Explanation: The balance after 3 years is $950(1.03)^3 = 1,038.14$, rounded to the nearest cent.

GED-MATH14-036 — Correct: C — Correct Answer: 7 mL — Explanation: The concentration is $300 \div 4 = 75 \text{ mg/mL}$. A 525 mg dose needs $525 \div 75 = 7 \text{ mL}$.

GED-MATH14-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: $T1=4x + 9y = 73$; $T2=x + y = 73$; $T3=y = x - 5$ — Explanation: Each tile matches the relationship described in its target.

GED-MATH14-038 — Correct: C — Correct Answer: 120 — Explanation: Oven A bakes $36 \div 24 = 1.5$ rolls per minute. Oven B bakes $45 \div 30 = 1.5$ rolls per minute. Together they bake 3 rolls per minute, so in 40 minutes they bake 120 rolls.

GED-MATH14-039 — Correct: B — Correct Answer: 17 — Explanation: The shelf must satisfy $110 + 48c \leq 950$. Then $48c \leq 840$, so $c \leq 17.5$. The greatest whole number is 17.

GED-MATH14-040 — Correct: B — Correct Answer: 2,475 — Explanation: After 2 seasons, the population is $2,800(0.94)^2 = 2,474.92$, which is closest to 2,475.

GED-MATH14-041 — Correct: A,B — Correct Answer: The total cost was \$186.30. | The boards cost \$103.50 in total. — Explanation: Boards cost $18(5.75) = 103.50$. Brackets cost $24(1.35) = 32.40$. Screws cost $6(8.40) = 50.40$. Total cost is 186.30.

GED-MATH14-042 — Correct: B — Correct Answer: 11 — Explanation: $f(8) = 0.25(8^2) - 2(8) + 11 = 16 - 16 + 11 = 11$.

GED-MATH14-043 — Correct: A — Correct Answer: $y = 2x^2 - 2x + 4$ — Explanation: Substituting $x = -2, -1, 0$, and 1 into $y = 2x^2 - 2x + 4$ gives 16, 8, 4, and 4? Correction: The listed table does not match choice A.

GED-MATH14-044 — Correct: A — Correct Answer: $24/(x - 3) - 6$ — Explanation: The difference of x and 3 is $x - 3$. The quotient of 24 and that difference is $24/(x - 3)$, and 6 less than that is $24/(x - 3) - 6$.

GED-MATH14-045 — Correct: 18 — Tolerance: exact — Correct Answer: 18 — Explanation: $301.44 = 1/3(3.14)(4^2)h = 16.746...h$, so $h = 18$.

GED-MATH14-046 — Correct: C — Correct Answer: 1,014 square feet — Explanation: Hall area is $42 \times 28 = 1,176$ square feet. Each square display area is $9 \times 9 = 81$ square feet, so two displays take 162 square feet. Remaining area is $1,176 - 162 = 1,014$ square feet.